

VITALS: Validity Intervals and Termination Attribution for Long-horizon Simulation

Working paper title: *Survival Analysis of Physical Coherence in Long-Horizon World-Model Rollouts*

Abstract

World models are increasingly deployed as predictive simulators for robotics, autonomy, and embodied agents. Existing benchmarks evaluate them at a *fixed* horizon and report a *scalar* quality score — physical plausibility, visual fidelity, instruction adherence, or a VQA-derived aggregate. Two problems follow. First, these designs cannot answer the question a practitioner actually has: *how long can I trust this model, and what breaks first?* Second, and more fundamentally, their task selection is unprincipled — nothing explains why a benchmark contains the scenarios it contains rather than any other set.

We address both by starting from a theory. We define **behavioral world understanding** as the capacity to maintain a coherent predictive model of the evolving physical world, formalize that capacity as a *structured stochastic transition model over a factorized state*, and derive from it **six observable properties that must necessarily hold** of any system possessing it: persistent frame structure, persistent entity representations, persistent relational representations, persistent physical dynamics, persistent causal consistency, and persistent uncertainty calibration. Each property is a necessary consequence of one component of the formal commitment; drop any component and the object is no longer a structured stochastic transition model.

VITALS is the benchmark this theory generates. Every element is derived rather than curated: each property yields an observable consequence, each consequence yields an episode family in which the property is load-bearing, each episode family yields an intervention family that stresses it, and each intervention yields a falsification test. Episodes that do not derive from a property do not enter the corpus, and gaps in the property $\times$ intervention matrix are reported as gaps.

Measurement is architecture-agnostic and fully quantitative. We make an auditable design commitment: **no language model appears anywhere in the measurement path**. Video is mapped to structured physical state — instance identities, poses, trajectories, contact and support relations, static scene geometry — by a specified geometric pipeline whose every component is validated against ground truth and whose error floors are published. Every episode is paired with a **reference ensemble** of independent realizations under a calibrated perturbation model, supplying a horizon-resolved noise floor that separates model error from intrinsic process variability. All detection thresholds are estimated as quantiles of statistics computed *between reference realizations*, so each detector carries a stated false-positive rate and is self-immunized against its own systematic weaknesses. The theory’s dependency ordering supplies the competing-risks precedence rule, which in prior designs was an arbitrary choice.

The corpus is generated in controlled simulation, which the theory forces rather than merely permits: property 5 requires exact branch replay, which no physical capture can provide. A small real-data arm anchors the perturbation scale in physical units and bounds the design’s principal residual risk, input-domain shift.

The theory produces three measurement modalities, and the primary results follow them: a **validity interval** and **termination profile** from per-rollout competing-risks survival analysis (P1–P4), a **counterfactual horizon** from paired factual/counterfactual branches (P5), and a **calibration horizon** from ensemble dispersion diagnostics (P6). VITALS reports no aggregate leaderboard scalar.

1. Motivation

1.1 Benchmarks currently have no theory of task selection

Ask why a physical-reasoning benchmark contains collisions, occlusions, and falling objects, and the honest answer is usually that those are the scenarios people build. Task selection is curated, not derived. This is the deeper form of the construct-validity problem: without an account of *what a system must be able to do*, there is no argument that the tasks measure the construct rather than a convenient sample of it.

VITALS begins from the opposite end. We state what a coherent predictive model of the physical world formally *is*, derive what must necessarily follow from it behaviorally, and let those consequences determine the corpus. This makes task selection falsifiable: one can argue that a derived property is not in fact necessary, or that a proposed episode does not in fact test it. Neither argument is available against a curated benchmark.

1.2 The practitioner’s question is a duration, not a score

A lab deciding whether to use a world model for closed-loop policy evaluation, model-predictive control, or synthetic data generation needs to know the horizon over which rollouts can be treated as evidence. That is a duration. Current benchmarks return a number between 0 and 1 at a horizon the benchmark chose.

Where long-horizon degradation is measured, it is measured as *quality decay* — drift metrics, progressive penalties on later frames, revisit consistency. These still produce a scalar at a fixed endpoint. They do not report *when* the rollout stopped being a valid prediction, and they do not distinguish a model that degrades into honest uncertainty from one that confidently hallucinates a coherent but wrong world.

1.3 Long-horizon evaluation is unidentifiable without a reference ensemble

Physical futures are stochastic. A single ground-truth continuation cannot distinguish “the model got the physics wrong” from “the model sampled a different valid future.” The standard workaround — sampling N futures and scoring the best match — is counterproductive: it rewards diffuse, high-variance sampling, the opposite of the property being measured. Distributional metrics such as FVD measure realism and diversity without reference to whether the *true* future was predicted, and independent analysis has documented their unreliability as standalone measures.

This compounds with horizon. As T grows, the entropy of the true future grows too, so any degradation curve over horizon confounds model degradation with irreducible process uncertainty. **A benchmark that does not measure the second cannot interpret the first.** The only demonstrated solution is an empirical noise ceiling obtained from repeated realizations of the same scenario — a design used by Physics-IQ at a single fixed horizon, which VITALS generalizes to a horizon-resolved reference distribution scored with a proper scoring rule. The requirement is structural: every episode needs multiple independent realizations of the same initial condition, at every horizon of interest.

1.4 The measurement methodology already exists in a neighboring field

Evaluating long-horizon predictions of a chaotic physical system when only one realization of truth is observed is the defining problem of operational numerical weather prediction, which has fifty years of mature verification theory. Its instruments — proper scoring rules (CRPS and multivariate extensions), skill scores against a reference, spread-skill ratios, rank histograms, and headline metrics expressed as *the lead time at which a skill threshold is crossed* — map directly onto world-model evaluation. To our knowledge this has not been applied to video world models.

1.5 A benchmark’s own measurement quality is currently unexamined

Across the world-model benchmark landscape we find almost no reporting of test-retest reliability, seed variance, confidence intervals, minimum detectable difference, or instrument error floors. Benchmarks relying on learned perception stacks or MLLM judges do not characterize the error of those instruments. §9 is our response.

2. Theory of Behavioral World Understanding

2.1 Definition

Behavioral world understanding is the capacity to maintain a coherent predictive model of the evolving physical world.

This is not directly observable. To make it measurable we commit to a formalization and derive observable consequences from it.

2.2 The formal commitment

A coherent predictive model of the physical world is a **structured stochastic transition model over a factorized state**.

Unpacked, this asserts six things, no more and no fewer:

$$p(s_{t+1} | s_t, a_t), \quad s_t = \left(\underbrace{\phi}_{\text{invariant frame}}, \underbrace{\{e_t^{(i)}\}}_{\text{entities}}, \underbrace{\{\rho_t^{(ij)}\}}_{\text{relations}} \right)$$

1. There is a **time-invariant substrate** ϕ — scene geometry and the observer–world relation — against which change is defined.
2. The time-varying part of the state is **individuated** into persistent entities $e^{(i)}$ with continuing identity.
3. The state **factorizes over relations** $\rho^{(ij)}$ between entities.
4. There is a **time-invariant transition operator** carrying s_t to s_{t+1} .
5. The transition operator is **modular**: it decomposes into mechanisms that can be independently intervened upon without altering the others.
6. The transition operator is a **distribution**, not a point estimate.

2.3 The six necessary consequences

#	Formal component	Necessary observable property
P1	Time-invariant substrate ϕ	Persistent frame structure — scene geometry and the observer–world relation remain consistent; what does not change, does not change
P2	Individuation into entities	Persistent entity representations — objects continue to exist and retain identity through occlusion, motion, and time
P3	Factorization over relations	Persistent relational representations — support, containment, contact, and adjacency remain coherent as the scene evolves
P4	Time-invariant transition operator	Persistent physical dynamics — motion predictions remain physically plausible over extended horizons
P5	Modularity of mechanism	Persistent causal consistency — interventions produce correct downstream effects and <i>only</i> those effects
P6	Distributional output	Persistent uncertainty calibration — predictive uncertainty tracks genuine physical ambiguity

Why exactly six. The six are the constituents of the formal object. Remove any one and what remains is not a structured stochastic transition model: without P1 there is no frame in which motion is defined; without P2 there is no state to factorize; without P3 the state is a bag of independent objects; without P4 there is no prediction; without P5 there is a lookup table rather than a model; without P6 there is a guess rather than a prediction. Any proposed seventh

property reduces to a consequence of these. This is the argument that makes “necessarily” do work rather than serve as decoration, and it is the claim a reviewer should attack if they wish to attack the paper.

2.4 The dependency ordering

P1–P4 form a strict dependency chain in which each is a precondition for the next being *well-defined*:

$$P1 \text{ (frame)} \prec P2 \text{ (entities)} \prec P3 \text{ (relations)} \prec P4 \text{ (dynamics)}$$

Motion is only defined relative to a frame, so P1 precedes everything. Relations hold between individuated, identified entities, so P2 precedes P3. Dynamics evolve a structured state, so P3 precedes P4.

This ordering is not decorative: it **derives the precedence rule** for assigning a single first cause when failures co-occur (§8.4), which in prior designs was an arbitrary stipulation and a named threat to validity. P5 and P6 sit outside the chain because they are properties of the *operator* rather than of the state, which is why they require different measurement modalities.

2.5 Three measurement modalities, derived

Properties	Property of	Measurement modality
P1–P4	An individual predicted trajectory	Per-rollout terminal risks → competing-risks survival analysis → <i>validity interval, termination profile</i>
P5	A pair of trajectories (factual, counterfactual)	Paired-branch divergence → <i>counterfactual horizon</i>
P6	The predictive <i>distribution</i>	Ensemble dispersion → <i>calibration horizon</i>

This asymmetry is a finding, not an inconvenience. A single rollout cannot be miscalibrated; only an ensemble can. A single rollout cannot violate mechanism modularity; only a matched pair can reveal it. The theory therefore predicts that a complete evaluation requires three distinct experimental designs, and a benchmark using only one is necessarily incomplete. No existing benchmark uses all three.

2.6 What the theory licenses – and what it does not

The theory establishes P1–P6 as **necessary conditions**. The only valid inference is the contrapositive:

┃ A model that fails any property does not possess behavioral world understanding.

The converse does **not** follow. Passing all six is not sufficient, and VITALS makes no claim that it is. Inferring understanding from passing behavior would affirm the consequent, the failure mode documented at length in the benchmark-validity literature.

VITALS is therefore a **systematic falsification battery**. Its contribution over a scalar benchmark is that failures are *localized* to a specific necessary condition and *timed*, rather than aggregated into a score. Its claim is not that a passing model understands the world; it is that a failing model demonstrably does not, and that knowing which condition failed and when is the actionable information.

Discriminant validity is therefore a required experiment (§5.4), not an appendix. If a nearest-neighbor retrieval baseline passes the battery, the items do not test the properties and must be redesigned.

2.7 Relation to causal representation learning

P5 adopts an existing formalization rather than inventing one: the **principle of independent causal mechanisms**, under which the conditional distribution of each variable given its causes neither informs nor influences the other

conditional distributions, so a mechanism remains invariant when other mechanisms change (Schölkopf et al., 2012; Peters et al., 2017; Schölkopf et al., 2021).

This yields the bidirectional test structure used throughout: an intervention must produce its downstream effects (**under-response failure**) and must not alter mechanisms it does not touch (**contamination failure**). The second direction is the one prior world-model benchmarks systematically omit.

The paper must cite and differentiate from the causal representation learning literature explicitly. The differentiation is clean: **that literature concerns learning and identifying causal representations; VITALS verifies behavioral consequences with no representational access whatsoever**. We assume nothing about a model’s internals and require nothing to be exposed. This is what makes the protocol architecture-agnostic and immediately deployable.

3. Derived benchmark design

Property → Observable consequence → Episode family → Intervention family → Falsification metric

3.1 The derivation table

Property	Observable consequence	Episode family	Intervention family	Falsifying measurement
P1 Frame structure	Static scene structure and the observer–world relation survive viewpoint change and revisit	Camera orbit and revisit over a static scene; navigation loop closure	Camera path perturbation; viewpoint jump; frame occlusion	Static-keypoint reprojection consistency; revisit mismatch
P2 Entity representations	Object identity and existence survive disappearance	Occlusion transit; multi-object tracking; container occlusion	Insert occluder; extend occlusion duration; add distractors	Instance-count deviation; identity continuity under reappearance
P3 Relational representations	Support, containment, and contact relations remain consistent	Stacking; containment; contact chains; nested assemblies	Remove supporting element; tilt substrate; break containment	Support-graph edit distance; interpenetration
P4 Physical dynamics	Motion predictions remain physically plausible	Rolling, bouncing, projectile, articulated linkage, collision	Perturb friction, mass, restitution, gravity; alter incline	Trajectory skill against reference ensemble
P5 Causal consistency	Interventions produce downstream effects, and only those effects	Domino chain; lever and pulley; pour and cascade; multi-stage mechanism	Remove one chain element; block one path; act on an unrelated object	Counterfactual branch divergence (under-response) and unaffected-region divergence (contamination)
P6 Uncertainty calibration	Predictive uncertainty reflects genuine ambiguity	Dice; spinning coin; unstable equilibrium; granular scatter	Modulate entropy — increase initial-condition sensitivity, tighten or loosen perturbation	Spread–skill ratio; rank-histogram/PIT uniformity

Every row is a falsification test of one necessary consequence. Every cell in the corpus traces to a row.

3.2 Intervention taxonomy, derived

State interventions change the underlying physical state. Each targets one property (column 4 above). Correct behavior is a *specific change* in the prediction.

Observation interventions change only the rendering — viewpoint, lighting, appearance, texture, distractor presence — leaving the physical state identical. Correct behavior is **invariance across all six properties**. This gives the ORBIT-

era “robustness to nuisance variation” a principled home: not a seventh property, but the *invariance requirement* attached to every property, and testable as such.

Two error types follow, reported separately: failing to respond to a state intervention, and responding to an observation intervention.

3.3 Completeness commitments

The theory is decorative unless it forbids things. Two falsifiable commitments, both stated in the paper and auditable against the released manifest:

1. **No episode enters the corpus unless it derives from a property.**
 2. **The property × intervention design matrix must be complete**, with empty cells reported as gaps and explained, not silently omitted.
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4. Research questions

RQ1 – Validity interval. What is the validity interval of current world models against a calibrated reference ensemble, and how does it compare to the intrinsic predictability of the same scenario?

RQ2 – Termination profile. Which necessary property fails first, and does the profile differ systematically across model families (diffusion vs. autoregressive; action-conditioned vs. passive; short- vs. long-context)?

RQ3 – Hazard modulation. How do controlled difficulty factors modulate each property-specific hazard? Reported as hazard ratios with confidence intervals.

RQ4 – Causal consistency (P5). Do models respond correctly to state interventions, and do they leave unaffected mechanisms unaltered? *Hypothesis: contamination failure is widespread and under-reported, because prior benchmarks test only the first direction.*

RQ5 – Uncertainty calibration (P6). Are world-model ensembles calibrated with respect to physical ambiguity? *Hypothesis: current models are severely under-dispersive at long horizons, producing confident single futures where the physics is genuinely multi-modal.*

RQ6 – Property independence. Are the six properties empirically separable, or do they collapse onto fewer dimensions? Factor analysis over per-property horizons is a direct test of the theory’s decomposition and can falsify it. *This is what makes the theory scientific rather than stipulative.*

RQ7 – Reliability and criterion validity. Is the ranking stable across seeds, conditioning modes, and resolutions? Does the validity interval predict closed-loop task outcome better than existing fidelity scores?

RQ8 – External validity. Does the ranking obtained in simulation transfer to real deployment distributions?

4A. Central claim

How long can a world model be trusted, and what kills it first?

A world model’s **physical coherence** on a scenario is (i) the duration over which its rollouts remain statistically indistinguishable from the intrinsic variability of the reference process, and (ii) the distribution over causes of first failure once that duration is exceeded.

A **good** world model has a long validity interval relative to the scenario’s intrinsic predictability, *and* a termination profile dominated by kinematic failure — meaning that when it fails, it fails by honest trajectory divergence rather than by objects vanishing, interpenetrating, or mutating. A **bad** world model has a short validity interval, or one dominated

by frame, existence, identity, or constraint failure, indicating that the rollout stopped depicting a physically coherent world at all.

Rollout termination is an observable event with a time and a first cause. The distribution of those times and causes, measured against a calibrated reference ensemble and instrumented with detectors of known error rate, is a valid, reliable, and architecture-agnostic characterization of a world model’s usable predictive range. VITALS is therefore a **falsification and characterization instrument, not a representational probe** (§2.6).

5. Positioning, claims, and validity

5.1 Already occupied — not claimed as novel

- *Long-horizon stability and drift*. WorldRoamBench, PAN, DreamX-World, and the autoregressive-drift literature.
- *Controlled interventions in simulation*. CRONOS (viewpoint, scene, object category, appearance held against fixed physical event type), What-If World (paired instances, one variable changed per pair).
- *Violation-of-expectation over core principles*. IntPhys and IntPhys 2 — the closest prior art to §2; the comparison must be explicit and generous.
- *Unified multi-dimension suites*. PAI-Bench, WorldScore, WorldArena, EWMBench, WorldModelBench, RoboWM-Bench.
- *Variance-normalized physical prediction*. Physics-IQ.
- *Causal representation desiderata*. Schölkopf et al. and the ICM literature (§2.7).
- *Perception components*. SAM 3, CoTracker3, DINOv2/v3, SfM — used off the shelf, not claimed.

5.2 Claimed as novel

1. **A derived rather than curated benchmark**, with completeness commitments a reviewer can audit.
2. **Six properties derived from one formalization**, with an argument for why exactly six.
3. **A dependency ordering that derives the competing-risks precedence rule**.
4. **Three measurement modalities derived from the theory**, with the claim that any benchmark using fewer is structurally incomplete.
5. **Time-to-event as the unit of measurement** — validity, counterfactual, and calibration horizons, all durations with confidence intervals.
6. **Competing-risks failure attribution**, converting attribution into an estimable statistical object.
7. **Reference-derived thresholds with stated false-positive rates**, removing threshold choice as a free parameter and self-immunizing detectors against systematic instrument bias.
8. **Bidirectional causal testing** — under-response *and* contamination.
9. **Ensemble verification methodology imported from NWP**.
10. **Perturbation-magnitude sensitivity as a reported result** (§7.4).
11. **A fully specified, language-model-free measurement operator with published per-component error floors** (§9), including an artifact-injected instrument validation set that does not currently exist.

5.3 Explicit non-claims

No claim about internal representations (§2.6). No claim that passing implies understanding. No claim of external validity to real deployment distributions in v1 (§14, T3).

5.4 Discriminant validity — a required experiment

Baseline	Expected behavior	What it tests
Copy-last-frame	Immediate P4 failure, no P2 failure	Detector sensitivity to motion
Static-scene	As above	Floor
Reference-vs-reference (leave-one-out)	Terminates at rate α by construction	Instrument false-positive floor
Ground-truth simulator rollout, matched perturbation	Ceiling	Upper reference
Nearest-neighbor retrieval from corpus	Should fail early on held-out initial conditions	Memorization shortcut
Property-ablated oracle (e.g. correct dynamics, randomized identity)	Should fail <i>only</i> the ablated property	Detector specificity — the sharpest test that the six measurements are separable

6. Design decision: simulation-first, real-anchored

The corpus is generated in controlled simulation. A small real-data arm anchors it. No component of v1 depends on new real-world capture, but §10.6 specifies the collection that would materially strengthen the work and should be sequenced as a parallel asset.

6.1 Why the theory forces simulation

P5 is uncomputable without exact branch replay. Causal consistency requires matched factual and counterfactual branches sharing an exact prefix and seed, differing in one intervention. This is physically impossible to capture at any budget. The design decision is therefore *forced by the theory*, not chosen for convenience — a stronger position than the usual cost argument.

Confound isolation. The diagnostic ambition is to say which factor caused which failure, which requires factors to vary independently. In real footage, camera motion, lighting, occlusion, complexity, appearance, and compression covary and attribution is not identified. In simulation every covariate varies one at a time with all others held exactly fixed. Decisive for RQ3.

Ground-truth relational and geometric structure. P3 detectors require the true contact/support graph; P1 detectors require true camera extrinsics and static geometry. These exist only in simulation. Without them the detectors can be applied but never validated — and §9.3’s relation head cannot be trained at all.

Scale and extensibility. The corpus regenerates with new scenarios, covariate levels, and longer horizons without re-collection.

6.2 The costs, stated honestly

Cost 1 — the reference spread is a design parameter. Deterministic simulators have no intrinsic variability; reference ensembles are produced by *injecting* perturbations, so spread is a hyperparameter. Thresholds are quantiles of that spread and horizons are functions of those thresholds. Controlled perturbation makes this arbitrariness *explicit*, not absent. Mitigation: mandatory λ sweep (§7.4, §11.5), affordable only because the corpus is simulated; optional absolute anchoring via §10.6.

Cost 2 — input-domain shift. Models receive a rendered conditioning prefix. If a model degrades on rendered input for non-physical reasons, VITALS partly measures render-domain robustness. Treated as a controlled covariate (§10.4) and bounded by the real arm (§10.6). Highest residual risk in the design.

Cost 3 — external validity unestablished by construction. RQ8, scoped to Phase IV.

Cost 4 — instrument validation on synthetic imagery. Mitigated with off-the-shelf real annotated data (§9.6) and the real arm.

Cost 5 — the simulator’s physics is itself a model. Contact solvers, restitution, and friction approximations become ground truth (§14, T4).

7. Formalization

7.1 The episode

An episode is the experimental unit — a controlled experiment with replication, not a video clip.

$$E = (s_0, \Sigma, C, A, \Xi, \mathcal{R}, T_{\max}, \pi^*)$$

Symbol	Meaning
s_0	Nominal initial configuration (frame, entities, relations, materials, camera, lighting)
Σ	Perturbation model over s_0 with scale parameter λ (§7.4)
C	Conditioning prefix, $ C = T_c$ frames
$A = a_{1:T}$	Action sequence; optional
$\Xi = \{(t_i, \xi_i)\}$	Intervention schedule, typed as state or observation (§3.2)
$\mathcal{R} = \{r^{(1)}, \dots, r^{(M)}\}$	Reference ensemble: M realizations $s_0^{(m)} \sim \Sigma(s_0; \lambda)$ simulated under (A, Ξ)
$T_{\max}$	Rollout duration
π^*	Target property the episode is designed to falsify — the link back to §3.1

The declared target property is what makes the corpus auditable: every episode states which necessary consequence it tests. For P5 episodes the unit is a **matched pair** (E, E') sharing s_0 , seed, prefix, and reference ensemble, differing only in Ξ .

An episode without a reference ensemble is not a VITALS episode.

7.2 Model output

Given (C, A, Ξ) the model produces $\mathcal{P} = \{p^{(1)}, \dots, p^{(N)}\}$, N independently sampled rollouts. Deterministic models yield degenerate ensembles; CRPS reduces to mean absolute error, reported rather than hidden.

7.3 The measurement operator (specified in §9)

$$\Phi : \text{video} \rightarrow z_{1:T}, \quad z_t = (\kappa_t, \mathcal{E}_t, \{c_t^{(e)}\}, \{m_t^{(e)}\}, \{d_t^{(e)}\}, G_t)$$

with κ_t static-scene keypoints and camera geometry (P1), $\mathcal{E}_t$ the identified entity set (P2), c, m, d poses, masks and appearance descriptors, and G_t the contact/support graph (P3). The output is organized by property, deliberately.

Two invariants:

1. **Symmetry.** Φ is applied identically to reference and model video. **The reference ensemble is measured through Φ from rendered frames, never read from simulator state.** Simulator ground truth is used only to *validate* Φ

and the detectors (§9.6), never to score. Violating this gives the reference an instrument advantage and inflates every horizon in the paper.

2. **Characterized error.** Every component has a published error floor from independently annotated data (§9.6).

7.4 Perturbation model, thresholds, and the sensitivity sweep

For each risk k , a deviation statistic $\sigma_k(\Phi(p), \{\Phi(r)\})_t$. **Thresholds are estimated, not chosen.** Let σ_k^{ref} be the same statistic computed leave-one-out between reference realizations:

$$\theta_k(\lambda) = \text{Quantile}_{1-\alpha}(\sigma_k^{\text{ref}}), \quad \alpha = 0.01$$

A genuine reference realization is falsely declared failed at rate α by construction. Thresholds are estimated **per scenario and per time bin**, because both process variability and instrument error grow with horizon.

This construction does more than avoid an arbitrary constant. Because the reference realizations contain the same objects, lighting, and visual difficulties as the candidate, **any systematic weakness of Φ is already inside the reference distribution and the threshold sits above it.** A detector fires only when the model’s deviation exceeds the deviation the instrument produces on ground truth itself (§9.5).

The λ problem and its resolution. Since θ_k depends on λ , so does every horizon. Rather than fix λ :

1. Define Σ with an interpretable scale — per-object position, orientation, velocity, mass, friction — all scaled by dimensionless λ .
2. Evaluate at $\lambda \in \{\lambda_0/4, \lambda_0/2, \lambda_0, 2\lambda_0, 4\lambda_0\}$.
3. Report all horizons, profiles, and rankings as functions of λ .
4. **Ranking invariance is a stated acceptance criterion.** Stable rank order establishes comparative conclusions independently of λ . Instability is a negative validity finding, published as such rather than resolved by selecting a favorable λ .

Sensitivity by quantity: raw horizons **high**; normalized horizons moderate; termination profiles low–moderate; hazard ratios **low**; calibration diagnostics **high**; ranking to be established empirically.

Absolute anchoring (via §10.6): λ_0 fitted so simulated reference spread matches observed real spread on matched scenarios at matched horizons, upgrading raw horizons from simulator-calibrated units to physically anchored seconds. If unavailable, the paper states plainly that horizons are in simulator-calibrated units — comparable across models, not interpretable as absolute physical durations.

8. Risks, detectors, and the survival object

8.1 Property-indexed terminal risks (P1–P4)

The ontology decomposes **ways a rollout dies**, not capabilities a model has. The risks are exhaustive (a rollout either survives to T_{max} or terminates from exactly one cause), **mutually exclusive** under the precedence rule of §8.4, individually detectable by a geometric pipeline, and directly estimable as cause-specific hazards. Each is indexed to the property it falsifies.

Risk	Property	Statistic σ_k	Validated against
R1 Frame	P1	Static-keypoint reprojection inconsistency; revisit mismatch on return to a prior viewpoint	Ground-truth camera extrinsics, static geometry
R2 Existence	P2	Per-class instance-count deviation from the reference distribution; track death not attributable to occlusion, absent	Ground-truth object presence and visibility

Risk	Property	Statistic σ_k	Validated against
		from every reference realization	intervals
R3 Identity	P2	Appearance-descriptor distance from t_0 conditional on a surviving track, against the within-reference drift distribution	Ground-truth object attributes
R4 Constraint	P3	Interpenetration (3D volume overlap, or mask overlap with depth ordering violation); support-graph edit distance from the reference edge set	Ground-truth contact events and support relations
R5 Kinematic	P4	Negative skill against the reference ensemble on 6-DoF pose / point-track trajectories, fair CRPS variant	Ground-truth object poses

Propriety caveat, stated explicitly: CRPS averaged componentwise across time, space, and channels is proper for the **marginals**, not the joint. For multi-entity relational quantities use the **Energy Score** or the variogram score.

8.2 Terminal events and censoring

With persistence requirement π consecutive frames:

$$T_k(p) = \min\{t : \sigma_k(p)_{t'} > \theta_k \ \forall t' \in [t, t + \pi]\}, \quad T(p) = \min_k T_k(p), \quad \delta(p) = \arg \min_k T_k(p)$$

If no risk fires by $T_{\max}$, the rollout is **right-censored**. The censoring fraction is a headline number: a benchmark most models survive has saturated.

8.3 Competing-risks estimation

Cause-specific hazard $h_k(t)$ and cumulative incidence $F_k(t) = \int_0^t h_k(u) S(u) du$, with $S(t) = \exp\{-\sum_k \int_0^t h_k\}$. Estimation via Aalen–Johansen (F_k), Kaplan–Meier (S), Gray’s test (between-model CIF comparison), cause-specific Cox regression with Fine–Gray as robustness check. Implementations: `scikit-survival`, `comprisk`, `lifelines`.

Treating competing events as ordinary censoring produces biased absolute-risk estimates. State this explicitly to preempt the reviewer who suggests five independent Kaplan–Meier curves.

8.4 The precedence rule — derived, not stipulated

When multiple risks cross threshold within the persistence window, the first cause follows the dependency ordering of §2.4:

$$R1 \succ R2 \succ R3 \succ R4 \succ R5$$

A violation at a lower level makes measurement at any higher level ill-defined. If the frame is broken, “the object moved” has no referent. If the object does not exist, its support relation has no referent. This removes what was a named threat to validity in prior versions.

8.5 P5 — counterfactual horizon (paired-branch modality)

Measured on matched pairs (E, E') differing in one state intervention, sharing seed and prefix. Two failure directions, from ICM:

- **Under-response:** divergence between the model’s factual and counterfactual rollouts falls below the reference-derived branch-separation threshold — the model failed to register the intervention.
- **Contamination:** divergence appears in regions the intervention provably could not have reached within elapsed time (a causal-cone constraint computed conservatively from simulator state) exceeding the within-reference threshold — the model rewrote unrelated mechanisms.

Reported as a **counterfactual horizon** per direction, with the ratio of under-response to contamination failures as a diagnostic.

8.6 P6 — calibration horizon (ensemble modality)

Not a per-rollout event; a property of $\mathcal{P}$.

- **Spread-skill ratio** $SSR(t)$ = ensemble spread / ensemble-mean error; ≈ 1 indicates uncertainty tracking actual error.
- **Rank histogram / PIT uniformity**, reported as a deviation statistic with a confidence band, not eyeballed. U-shaped = under-dispersion; dome-shaped = over-dispersion.
- **Calibration horizon** = the time at which uniformity is first rejected at level α , or SSR first leaves a reference-derived band.

9. The measurement operator, concretely

This section is the answer to the question every reviewer will ask of a pixel-space benchmark: *what, exactly, is being extracted from the video?*

9.1 Design commitment: no language model in the measurement path

VITALS contains no language model anywhere in its measurement pipeline. No VLM judge, no MLLM scorer, no text-conditioned component, no language-supervised features. This is a stated, auditable commitment, not a preference.

The argument is not architectural — the pipeline contains neural networks throughout. It is about **error characterization**:

	Perception component (e.g. a segmenter)	Language-model judge
Output	A measurement (a mask, a track, a pose)	A verdict (a score)
Ground truth exists?	Yes — per-pixel instance IDs, poses, contact events from the simulator	No. There is no ground-truth “physical plausibility” scale anywhere
Error boundable?	Yes, and published to three decimals (\$9.6)	Not even in principle
Errors correlated with the models judged?	No	Yes — judges share training distributions with the systems they score

A segmenter produces a mask; *our code* then computes instance counts, overlaps, and trajectory deviations. All judgment is arithmetic we wrote and can inspect. A judge performs the judgment itself and offers no way to check it.

Two practical consequences for component selection:

- Where a component offers both a text pathway and a visual-prompt pathway, **the visual pathway is used**. Modern promptable segmentation accepts short noun phrases *or* image exemplars; we prompt with the ground-truth instance mask from frame 0, which the episode manifest supplies. This is a stronger prompt than text — zero semantic ambiguity — and it keeps the text encoder out of the loop.
- **Self-supervised visual features only**. DINO-family features are used for appearance descriptors; CLIP-family features are excluded because they are language-supervised. These are frequently conflated because both “produce features”; the distinction matters here.

9.2 Architecture: verification, not open-world perception

Φ never asks *what is in this video*. The episode manifest declares object count, classes, initial poses, materials, and camera parameters. Φ asks only: *are the objects I know should be here still behaving like those objects?*

This collapses a hard problem into a tractable one. The operating regime is semi-supervised video object segmentation with first-frame prompts, not open-vocabulary discovery. It also means every component has a validation target that was generated, not annotated.

9.3 Component specification

Output	Extracted by	Feeds
Instance masks + persistent IDs	Promptable video segmentation (SAM 3 / SAM 2 class), prompted with ground-truth frame-0 masks	P2 (R2, R3)
Point trajectories + visibility flags	Joint point tracking (CoTracker3 / Track-On2 class), N points seeded per object from frame-0 masks	P4 (R5), P2 (R2)
6-DoF object pose	Kabsch/Procrustes rigid fit over ≥ 3 visible tracked points per rigid body	P4, P3
3D positions	Multi-view triangulation where available; otherwise monocular depth / feed-forward geometry	P3 (R4), P4
Appearance descriptors	DINO-family self-supervised features, mask-pooled	P2 (R3)
Static keypoints + camera pose	Learned keypoints + matcher (SuperPoint/LightGlue class) on non-instance pixels $\rightarrow$ PnP + bundle adjustment, or feed-forward SfM	P1 (R1)
Contact / support graph	Learned relation head — §9.4	P3 (R4)

Design notes, each load-bearing:

Visibility flags are what make P2 measurable. The entire question in an occlusion episode is whether a hidden object is treated as *hidden* or as *gone*. Point trackers predict visibility explicitly alongside position, so “occluded but tracked” and “track died” are distinct states rather than the same observation. R2 fires on the second, not the first.

Track points, not centroids. A mask centroid is biased the instant an object is partially occluded — it drifts toward whatever remains visible. Fitting a rigid transform over ≥ 3 tracked points yields true 6-DoF pose and degrades gracefully under partial occlusion. This makes R5 a pose statistic rather than a position statistic, which is strictly more informative.

Static/dynamic separation is free. Any pixel not covered by an instance mask is a candidate static point. These feed camera estimation; their reprojection error under the estimated camera is R1’s statistic.

Interpenetration should be 3D where possible. A 2D mask-overlap proxy confuses occlusion with interpenetration. With multi-view or reliable depth, R4 tests genuine volume overlap. Where only monocular video is available, mask overlap must be combined with depth ordering, and the weaker guarantee reported.

9.4 The relation head — the critical-path component

There is no off-the-shelf model that extracts a contact/support graph from video. This is the one genuinely open component and the highest-risk item in the project.

The solution turns the simulation-first design into an advantage: **train it on Tier S, where ground-truth contact and support graphs are emitted for free at unlimited scale.**

- **Inputs (all from Φ ’s other outputs, not raw pixels):** pairwise 3D surface proximity, mask-boundary adjacency, relative vertical displacement along gravity, relative velocity, contact persistence over a temporal window, and object mass/material from the manifest where the model under test was also given them.
- **Outputs:** per-pair edge labels over {contact, supported-by, contained-in, none}.

- **Architecture:** a small GNN or pairwise classifier — deliberately low-capacity, so it generalizes on geometry rather than memorizing scene appearance.
- **Training data:** hundreds of thousands of labeled frames from Tier S, free.
- **Validation:** held-out Tier S episodes (§9.6) plus the real anchor set (§10.6).

Two honest caveats. The head is trained on rendered video, so its real-domain error is unknown until checked — one more argument for the real arm. And **if the head does not reach acceptable precision, P3 becomes unmeasurable and the correct action is to report it as a gap rather than ship a weak detector**. This is Phase I’s principal technical risk and should be attacked first.

9.5 Instrument failure modes, and why they mostly cancel

A geometric pipeline has characteristic failure modes: a dark object against its own shadow, a specular highlight read as a second object, a thin occluder splitting a mask. Four mechanisms address them, in ascending order of importance.

1. **Persistence requirement.** Confusion is typically intermittent; requiring π sustained frames rejects transient tracker error. Weakest defense, free.
2. **Measured, not assumed.** §9.6 quantifies exactly how often each confusion occurs, per scenario, against ground-truth instance IDs. This does not fix the problem; it converts an unknown into a **bounded known** — which is the entire difference between this instrument and a judge.
3. **Symmetry cancels systematic bias.** Φ runs identically on reference and model video, and σ_k is a *deviation between* them. An instrument that confuses shadow for object does so on both sides; symmetric bias does not fire a detector.
4. **Reference-derived thresholds absorb the residual.** θ_k is a quantile of the statistic computed *between reference realizations*, which contain the same object, the same shadow, and the same confusion. Instrument-induced jitter is therefore already inside the reference distribution and the threshold sits above it. **A detector fires only when the model’s deviation exceeds the deviation the instrument produces on ground truth itself.**

Worked example — a dark ball rolling across its own shadow. The segmenter is prompted with the ball’s ground-truth frame-0 mask and carries temporal memory; shadows have soft boundaries, lie on the support surface, and do not match the exemplar. If the mask leaks anyway, tracked points seeded on the ball triangulate above the support plane while shadow points land on it — geometry separates what appearance could not, which is the strongest single argument for multi-view capture in §10.6. If confusion persists, it is symmetric and cancels (3). Residual asymmetry sits inside the threshold (4). And if the object is genuinely unrecoverable from the imagery, §9.6 reports low recall for that scenario, **the episode is excluded, and the exclusion rate is published**. A measurable instrument knows when it has failed; a judge would return a confident number either way.

The one place this argument has a hole. Reference video is a clean render; model video is generated and carries artifacts. Φ may behave differently on the two, and *that* asymmetry does not cancel. This is precisely what the artifact-injected validation set (§9.6) exists to bound, and it is the strongest reason not to defer that work.

9.6 Validation protocol, per component

Every component ships with a published error floor. Nothing enters the pipeline unvalidated.

Component	Validated against	Reported
Segmentation + identity	Tier S per-pixel instance IDs	Mask IoU, ID-switch rate, false-merge rate per scenario
Point tracking	Tier S ground-truth tracks; external real annotated tracks (TAP-Vid class)	Position error, occlusion-flag accuracy, sim–real gap

Component	Validated against	Reported
Pose fitting	Tier S object poses	Translation and rotation error vs. number of visible points
Camera + static geometry	Tier S extrinsics	Reprojection error, drift over horizon
Relation head	Held-out Tier S contact/support graphs	Per-edge-type precision, recall, event-time bias
Full detector chain	Held-out Tier S episodes with known event times	Per-risk precision, recall, event-time bias
Artifact robustness	Artifact-injected set — reference renders degraded with graded generation-style artifacts (blur, colour drift, temporal flicker, local warping)	Error of every component as a function of artifact severity
False-termination floor	Leave-one-out reference realizations	Per-risk false-termination rate, should approximate α

The artifact-injected set does not currently exist. Building and releasing it is a contribution independent of the benchmark, since it is the missing instrument for anyone measuring generated video geometrically.

10. Data

Everything follows from one requirement: **multiple independent realizations of the same initial condition, at every horizon of interest.**

10.1 Tier S — synthetic core (the benchmark)

Property	Specification
Episode families	6 (one per property, §3.1), each with 15–25 scenarios
Total scenarios	100–150
Reference realizations	$M = 30$ per scenario per λ level
λ levels	5
P5 pairing	Every P5 scenario generated as a matched factual/counterfactual pair sharing seed and prefix
Duration	60 s minimum at 24–30 fps
Ground truth exported	Camera extrinsics/intrinsics, static geometry, object poses, velocities, contact events, support graph, occlusion masks, per-pixel instance IDs, material parameters, causal-cone masks for P5
Determinism	Full seed control; bit-reproducible regeneration from the manifest
Rendering budget	Full corpus at mid realism (rasterized); path-tracing reserved for the realism-covariate subset only

Rendering cost is a design constraint, not an afterthought. At 150 scenarios $\times$ $M=30 \times 5$ lambda levels $\times$ 60 s $\times$ 30 fps the corpus is order 4×10^7 frames. Rasterized rendering at ~ 50 ms/frame is a few hundred GPU-hours and affordable; path-tracing at ~ 2 s/frame is not. The corpus is therefore generated at mid realism, with path-traced rendering applied only to the graded-realism subset in §10.4. Reducing to 100 scenarios and $M=25$ buys back roughly a third.

Predictability bands map onto episode families rather than forming an independent axis: P1 and P4 are predominantly Band I (near-deterministic), P3 and P5 Band II (moderately chaotic), P6 Band III (high-entropy) by construction. P2 spans bands via occlusion duration.

Simulator selection criteria, ranked:

1. **Exact branch replay** — fork a simulation at time t and apply an intervention with all else bit-identical. **Required for P5; the likely binding constraint and the first thing to test.**
2. Deterministic seeding **plus** parameterized perturbation of initial conditions and physical parameters (required for Σ).
3. Ground-truth export at the granularity above — especially contact events and support relations, without which §9.4 cannot be trained.
4. Programmatic mid-episode intervention API.
5. **Independent** control of observation-intervention factors — viewpoint, lighting, appearance, distractors — decoupled from each other. Most engines couple these; the criterion most likely to eliminate candidates.
6. Photorealistic rendering sufficient to keep models in-distribution (§10.4).
7. Multi-camera rendering, so the synthetic arm matches the real arm’s sensing (§10.7) and 3D interpenetration is directly measurable.
8. Throughput for $M=30 \times 5 \lambda_{\text{levels}} \times 100+$ scenarios $\times$ paired branches.

Candidate families and the current recommendation. MuJoCo / MJX is the working choice, on two grounds. First, criterion 1: all simulation state lives in `mjData`, so a fork is a struct copy and exact branch replay is native rather than a research problem. Second, availability — Isaac Sim requires Ubuntu or Windows with an RTX-class GPU and has no macOS build, so it is not viable on the current development hardware regardless of its other merits. Kubric (physics + Blender, purpose-built for ground-truth-annotated synthetic video) is worth evaluating as a render/export harness, potentially with its physics layer swapped for MuJoCo. Genesis and Unreal-based pipelines remain candidates for the photorealism criterion.

Decouple physics from rendering. Simulate once, record the state trajectory, then render that trajectory N times at different viewpoints, lighting, appearance, and realism levels. This makes observation interventions (§3.2) *free re-renders of identical physics* — guaranteed at the architecture level rather than argued for — and makes the render-realism covariate (§10.4) a controlled comparison with physics held exactly fixed.

Wrapper frameworks exposing controllable factors of variation over multiple backends are worth evaluating, since the observation-intervention design is a factors-of-variation problem.

Hard exclusion: generative models must not produce reference data — including as renderers. Using a learned world model to generate references would mean evaluating world models against a reference whose dynamics are themselves unvalidated, measuring agreement with another model rather than with physics. References must come from a physics engine with known dynamics, rendered by a physically-based renderer solving the rendering equation over analytic geometry.

The exclusion extends to the rendering layer, and the distinction that matters is **faithful vs. generative**, not learned vs. classical. A renderer that deterministically projects explicit scene state is admissible however it is implemented; a renderer that fills in plausible detail absent from the state is not, because it begins contributing physics the manifest never specified. Any candidate renderer must pass four tests before use: **faithfulness** (translate an object by δ in the state; the rendered object moves by exactly the projected δ , linearly, across the full λ range), **determinism** (identical state renders to identical pixels — required for P5 branch identity), **generalization** at $4\text{-}\lambda_0$, where configurations are furthest from any training distribution, and **exact camera control** with ground-truth passes. A renderer meeting these may enter as the top rung of the realism ladder (§10.4), where ranking invariance across rungs becomes a reportable result; it may not replace the ground-truth-emitting renderer on which detector validation (§10.2) and the relation head (§9.4) depend.

10.2 Detector and component validation sets (within Tier S)

Held-out episodes with full simulator ground truth, used to validate that Φ and the detectors — operating on *rendered frames* — recover the true events. Per §9.6. **The only place simulator state touches the pipeline**, used for validation, never scoring.

10.3 Instrument validation set (external, real, off the shelf)

Real annotated point tracks (TAP-Vid class) for the tracker error floor on real imagery, and the artifact-injected set of §9.6. Neither requires new capture.

10.4 Input-domain shift as a controlled variable

The sharpest threat to a simulation-first design.

1. **Render realism is a covariate.** A scenario subset generated at graded realism (flat-shaded → PBR → path-traced, with and without simulated sensor noise, motion blur, rolling shutter), included in the hazard model. A large hazard ratio here is a first-order finding bounding every other result.
2. **Domain-shift control.** Compare model rollout statistics from rendered vs. real prefixes of matched scenes (§10.6). A behavioral consistency check on the *models*, not a reference-ensemble requirement — a handful of scenes suffices.
3. **Report per-model, never pooled.** Models trained heavily on synthetic data will be flattered.

10.5 Excluded: web-scraped and found footage

Observational, not experimental: no repeated realizations, no interventions, nuisance confounded with content, unknown provenance, probable contamination. Only experimental data identifies counterfactual quantities without untestable assumptions — and P5 is uncomputable without exact branch replay, which found footage can never supply.

10.6 Tier R — the real anchor arm

Real data plays exactly three roles, with very different economics. All three are worth doing; only the third should be sequenced after v1.

Role	Scale	Value	Sequencing
λ_0 anchoring (§7.4) — fits the perturbation scale so horizons read in physical seconds rather than simulator-calibrated units	10–15 scenarios × 25 takes	High. Upgrades every headline number’s interpretability	v1. Best value-to-cost ratio in the project
Domain-shift control (§10.4) — rendered vs. real prefixes on matched scenes	~10 matched scenes, no repeats needed	High. T2 is the highest residual risk and this is the only thing that bounds it	v1. Cheap
Full real arm answering RQ8 — does the simulation ranking transfer?	40–60 scenarios × 25 takes, multi-view	Large. Converts v1’s stated limitation into a result	v2. Do not block the paper on it

Why the economics differ for a data-collection organization. A bespoke capture campaign is a prohibitive dependency for a small academic team, which is why v2 of this proposal excluded it. That reasoning does not transfer to an organization whose marginal-cost activity is collection. The resulting asset is also a durable moat: a real, repeated-trial, multi-view physics corpus with 25+ takes per scenario **does not exist publicly**. Physics-IQ is the nearest thing and it is small. Whoever holds that corpus holds the only credible external-validity check on synthetic world-model evaluation — which every lab publishing a synthetic benchmark eventually needs.

What real data cannot do, so as not to over-collect. P5 requires exact branch replay and is physically impossible to capture. A degraded version is available — run 30 takes with a domino present and 30 with it removed, and test whether the model predicts a different *distribution* (under-response). The contamination direction is unrecoverable, because

there is no matched pair to difference against. **Tier R therefore covers P1–P4 and P6, and P5 only partially. Scope collection accordingly.**

10.7 Collection specification (Tier R)

- **Scenarios:** mirror the Tier S episode families so the arms are comparable — that is the entire point. Prioritize P4 (dynamics) and P6 (uncertainty), where the sim–real gap is most likely to bite.
- **Takes:** $M \geq 25$. Below ~15 the reference distribution is too noisy at long horizons to estimate a 99th-percentile threshold.
- **Cameras:** 2–3 calibrated static views, rigidly mounted, plus depth where the scenario permits. Multi-view is not gold-plating: it makes R4 measurable as true 3D interpenetration rather than a 2D proxy, and it is the mechanism that defeats the shadow case of §9.5.
- **Fixed capture:** focus, exposure, white balance, ISO, frame rate. No auto-stabilization. Intra-frame codec or raw. Continuous flicker-free lighting.
- **Registration:** calibration target per session; fiducials in frame for metric scale. For visually difficult objects, markers on the objects themselves are cheap and standard practice in the deformable-tracking literature.
- **Reset discipline:** full scene reset between takes; never reuse a settled configuration. **Measure and log re-instantiation tolerance** — it is the resolution floor of the entire real arm and reviewers will ask.
- **Duration:** 3–5× the intended $T_{\max}$. The headline is a horizon; if models saturate the clip length, no horizon can be measured.
- **Licensing:** collected in-house, permissively released, no third parties, no identifiable persons. A real adoption factor — labs will not build on a corpus with murky provenance.
- **Exclusions for v1:** deformables (cloth has no persistent identity coordinates, no rigid pose, and changes topology under folding, so R3 and R5 are undefined on it) and egocentric capture (point tracking degrades substantially under fast camera motion, objects leaving and re-entering frame, and heavy occlusion — the same phenomena R1 and R2 detect, an adversarial correlation between instrument error and signal). Prefer fixed third-person streams.

Existing inventory. If prior collection work already contains clusters of near-identical real trials, these can serve the first two roles at no marginal cost. Mining: cluster by (station, object class, initial pose tolerance, action prefix), retain clusters with $M \geq 15$, validate tightness against Tier S spread on comparable scenarios.

11. Protocol and metrics

11.1 Episode construction

Phase 0 — Specification. Manifest before generation: target property π^* , s_0 , Σ and parameterization, episode family, intervention schedule Ξ with type, covariates including render realism, $T_{\max}$, M , λ grid.

Phase 1 — Reference generation. For each λ , sample M initial conditions from $\Sigma(s_0; \lambda)$, simulate under (A, Ξ) , render under identical camera and lighting. For P5, fork at t_i and generate the counterfactual branch with everything else identical. Export ground truth to a separate channel used only for validation.

Phase 2 — Conditioning. Prefix C drawn from a held-out realization **not** in $\mathcal{R}$, so the model is not scored against a distribution containing its own conditioning.

Phase 3 — Intervention. Apply Ξ at scheduled times, identically across all realizations, communicated to the model through whatever channel it supports. Channel recorded as a covariate.

Phase 4 — Model rollout. N rollouts per episode under a fixed, published input-normalization protocol: resolution, frame rate, prefix length, sampling/seed policy, conditioning modality. Comparison only within matched settings. *Existing benchmarks document that models produce wildly different displacement magnitudes for identical actions; without normalization, cross-model comparison is invalid.*

Phase 5 — Measurement. Φ applied symmetrically to $\mathcal{R}$ and $\mathcal{P}$, from rendered frames only (§7.3, §9).

Phase 6 — Threshold estimation. $\theta_k(\lambda)$ per scenario, per time bin, leave-one-out.

Phase 7 — Event extraction. T_k , precedence rule, (T, δ) or censoring. For P5, branch-divergence crossings. For P6, ensemble diagnostics. Episodes failing §9.6 recall thresholds are excluded and counted.

Phase 8 — Estimation and reporting.

11.2 Primary: Validity Interval (P1–P4)

$$VI_q(\lambda) = \inf\{t : \hat{S}(t) \leq q\}$$

Report VI_{50} (headline), VI_{90} (planning-relevant), bootstrap CIs, censoring fraction, exclusion rate, always with λ .

Normalized: $\widetilde{VI}_q = VI_q(\text{model})/VI_q(\text{reference-vs-reference})$; 1.0 means rollouts survive as long as a reference realization does.

Per-property horizons. Because episodes declare π^* , VI is also reported *per property*, restricted to episodes designed to test it. This is the direct answer to “which necessary condition does this model fail, and how soon.”

11.3 Primary: Termination Profile

Aalen–Johansen CIFs $(F_1, \dots, F_5)$ as full curves, summarized at $T_{\max}$ as a normalized five-vector, compared between models with Gray’s test. A profile dominated by F_5 (kinematic) indicates a model that maintains a coherent world and diverges honestly; one dominated by F_1 or F_2 indicates rollouts that stopped depicting a coherent world at all.

11.4 Primary: Counterfactual and calibration horizons

Per §8.5 and §8.6, with confidence intervals, reported alongside VI as co-equal primary results — one per measurement modality:

Output	Modality	Properties
Validity interval + termination profile	per-rollout competing risks	P1–P4
Counterfactual horizon (under-response / contamination)	paired factual/counterfactual branches	P5
Calibration horizon	ensemble dispersion	P6

A benchmark reporting only the first is measuring one of three necessary things (§2.5).

11.5 Primary: λ -sensitivity and ranking invariance

Horizons, profiles, and rank order across five λ levels with rank correlation between levels. **Acceptance criterion:** stable rank order establishes comparative conclusions independently of λ ; instability is a negative validity finding, published as such.

11.6 Secondary: Hazard ratios

Cause-specific Cox per risk. Covariates: object count, intervention type and magnitude, occlusion duration, predictability band, observation-intervention levels, compositional novelty, conditioning length, render realism, and a time-varying post-intervention indicator. Output: hazard ratios with CIs — “one additional distractor multiplies existence-failure hazard by 1.8 (95% CI 1.4–2.3).” Fine–Gray as robustness check.

11.7 Secondary: Property independence (RQ6)

Factor analysis over per-property horizons across models. If the six collapse onto fewer dimensions, the theory’s decomposition is empirically over-specified and the paper must say so. This is the analysis that makes §2 falsifiable.

11.8 Instrument floor (reported alongside every result)

Everything in the §9.6 table, plus per-scenario exclusion rates. Instrument floors are reported per tier and **never pooled**.

11.9 Design commitment: no aggregate scalar

Aggregating three horizons, a five-way termination profile, and calibration diagnostics into one number destroys the diagnostic content that motivates the benchmark and invites optimization against the metric. The reporting unit is the set. State this in the abstract; the answer is a feature.

12. Reliability and statistical protocol

- **Test–retest** across sampling seeds; rank correlation across runs.
- **Ensemble-size sensitivity** in N and M , justifying chosen values.
- **Internal consistency** across episodes within a family — also a check that the family coheres around its target property.
- **Minimum detectable difference** with a power calculation, so readers know when two models are tied.
- **Confidence intervals on everything**, bootstrapped over episodes (not rollouts) to respect clustering.
- **Pre-registration**: detector statistics, α , π , precedence rule (theory-derived), covariate set, λ grid, exclusion criteria, and the property $\times$ intervention matrix, frozen and published before any model is evaluated.
- **Regenerable held-out split** with undisclosed seeds — a structural advantage of a generative corpus over a fixed-asset benchmark.

13. Inherited from ORBIT and prior versions — kept, revised, dropped

Element	Disposition
“What observable behaviors necessarily emerge from a coherent predictive model?”	Restored and made central (§2), with a derivation rather than an assertion
Six-property ontology	Restored, reordered, re-derived — frame structure added as P1; nuisance robustness and compositionality demoted to invariance requirements and covariates (§3.2)
Necessity framing	Kept , with the sufficiency inference explicitly renounced (§2.6)
Representational claim	Dropped permanently
Episode as atomic unit	Kept , extended with declared target property π^* and matched-pair structure for P5
Five-level functional taxonomy	Dropped — replaced by the dependency ordering (§2.4), which does real work
Competing-risks survival machinery	Kept in full , now indexed by property
Arbitrary precedence rule	Replaced by a rule derived from the dependency ordering (§8.4)
Reference ensembles, estimated thresholds, λ sweep, instrument floors	Kept in full

Element	Disposition
“Calibration” as a name for degradation curves	Dropped and repurposed — dispersion calibration, elevated to a necessary property (P6)
Interventions as covariates only	Revised — interventions are the falsification instrument, typed state vs. observation, with P5 as their dedicated property
Aggregate scalar	Dropped
YouTube-scraped corpus	Dropped
Filmed corpus as the primary data source	Dropped — simulation is the corpus; real data is an anchoring arm (§10.6) with a stated, bounded role
Unspecified “geometric stack”	Replaced by the full component specification, validation protocol, and no-language-model commitment of §9

14. Threats to validity

T1 — The theory may be wrong. The six-fold decomposition is substantive and could be over- or under-specified. *Mitigation:* RQ6 factor analysis and the property-ablated oracle (§5.4). A negative result is publishable and should be published.

T2 — Input-domain shift. Models may degrade on rendered prefixes for non-physical reasons. *Mitigation:* §10.4 covariate design plus the §10.6 domain-shift control. *Residual risk:* highest in the design; belongs in the abstract’s limitations, not buried.

T3 — External validity. Ranking established in simulation; transfer unestablished. Scoped to RQ8 and Phase IV.

T4 — The simulator’s physics is itself a model. Contact solvers, restitution, and friction approximations become ground truth; a model penalized for disagreeing with a contact solver is not necessarily wrong. *Mitigation:* cross-engine agreement on a subset; engine identity as a covariate if two backends are viable. Under-discussed in the synthetic-benchmark literature; give it a paragraph.

T5 — Perturbation-magnitude arbitrariness. *Mitigation:* mandatory λ sweep with ranking invariance as acceptance criterion; absolute anchoring via §10.6.

T6 — Relation-head reliability. P3 depends on a bespoke component with no external baseline. *Mitigation:* trained and validated on unlimited Tier S ground truth; real-domain error checked against Tier R. *Residual risk:* if precision is inadequate, P3 is reported as an unmeasured gap rather than shipped weak. **Phase I’s principal technical risk.**

T7 — Instrument–signal correlation on generated video. Generation artifacts degrade Φ and simultaneously constitute the failure detected; symmetry does not cancel this because reference video is a clean render. *Mitigation:* the artifact-injected validation set bounds it. Report the bound; do not claim elimination.

T8 — Threshold sensitivity. α and π remain free. *Mitigation:* pre-register; publish an (α, π) sensitivity grid; report ranking instability if present.

T9 — Causal-cone specification for P5 contamination. Determining which regions an intervention could not have reached requires a defensible propagation bound. *Mitigation:* compute conservatively from simulator state and report the bound’s tightness; an over-tight cone manufactures contamination failures.

T10 — Ensemble sampling cost. $N \times \text{episodes} \times \lambda \text{ levels} \times \text{paired branches} \times \text{models}$ is a real budget. *Mitigation:* fair-CRPS permits smaller N ; the λ sweep can use a scenario subset.

T11 — Degenerate ensembles. Deterministic models make P6 undefined or trivial. Flag rather than hide.

T12 — Scope. V1 is rigid-body and articulated. No deformables, no fluids beyond coarse Band III, no optical or thermal phenomena.

T13 — Construct scope. VITALS measures the six necessary properties, not semantic appropriateness, instruction following, task success, or planning utility. RQ7 is the only bridge to the last, and it is correlational.

15. Phased plan

Phase I (months 0–3) — Theory and instrument.

Finalize the derivation (§2) and the design matrix (§3). Select and validate the simulator against §10.1 criteria — *exact branch replay for P5 is the binding constraint and should be tested in week one*. Build Φ per §9. **Train and validate the relation head (§9.4); this is the critical path.** Build the artifact-injected validation set. Publish all component error floors. *Deliverable:* a validated, language-model-free measurement pipeline with known error rates, and a pre-registered design matrix. No model results. Do not rush; every later claim depends on it.

Phase II (months 3–6) — Corpus.

Generate Tier S: six episode families, 100–150 scenarios, $M=30$, five λ levels, graded render realism, matched P5 pairs, multi-camera rendering. In parallel, collect the small Tier R anchoring and domain-shift sets (§10.6, §10.7). *Deliverable:* a released, regenerable corpus with manifests, seeds, target-property declarations, and reference ensembles; plus the λ_0 fit.

Phase III (months 6–9) — Evaluation.

Run 6–10 models plus the full baseline ladder including the property-ablated oracle. Produce per-property validity intervals, termination profiles, counterfactual and calibration horizons, hazard models, the λ sweep, RQ6 factor analysis, and the reliability suite. *Deliverable:* the paper.

Phase IV (post-paper) — External validity.

RQ8: the full Tier R arm (40–60 scenarios $\times$ 25 takes, multi-view), compared against the simulation ranking, plus a downstream closed-loop criterion.

Minimum viable result. If constrained: **three properties (P2 entity, P4 dynamics, P6 calibration) — one per measurement modality — with their episode and intervention families, three λ levels, the full baseline ladder, and complete reliability and instrument-floor analysis.** One property per modality preserves the structural claim of §2.5 while roughly halving corpus size and, critically, **deferring the relation head**, which removes the largest technical risk from the critical path.

16. Deliverables

1. The theory and derived design matrix, pre-registered.
 2. Released corpus generators, episode manifests with target-property declarations, and seeds — regenerable, not a fixed asset.
 3. Open-source harness: Φ with all components, the relation head and its weights, threshold estimation, survival/CIF estimation, branch-divergence and calibration diagnostics, hazard models.
 4. **The artifact-injected instrument validation set** and published per-component error floors — a standalone contribution for anyone measuring generated video geometrically.
 5. Tier R anchoring and domain-shift sets, permissively licensed.
 6. Paper: *Survival Analysis of Physical Coherence in Long-Horizon World-Model Rollouts*.
 7. Regenerable held-out split with undisclosed seeds, and a submission protocol.
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Selected references

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